

SEEING THE PRODUCT: A Multimodal AI Lens for Online Fashion

How an AI model could see a single product photo and instantly know what it is, what to call it, and where it belongs — inspired by Myntra's catalog and search experience

By Akash C Shetty

GenAI Bootcamp — Multimodal AI System Design Assignment

Case study reference: Myntra (India's largest fashion & lifestyle e-commerce platform)

In one line

A seller drops in one product photo; a vision-language model reads the image the way a human merchandiser would, and writes back the category and a ready-to-publish description — turning a five-minute manual task into a five-second one.

01 Inputs

At its heart, the system needs just one thing to get going — but a couple of optional signals make it noticeably smarter:

- **Primary:** a single product photograph — a seller's catalog shot, or even a casual phone photo of an item someone wants listed.
- **Helpful, not required:** a brand name or rough category typed by the seller, which the model can use as a soft hint rather than a hard rule.
- **Multiple angles (optional upgrade):** front, back, and close-up-of-fabric shots let the model describe texture and fit far more confidently than one photo alone.

02 Outputs

- **Product category:** placed correctly inside the catalog tree, e.g. Women > Western Wear > Bodycon Dresses.
- **Product description:** a short, shoppable paragraph — fabric, fit, occasion, styling notes — written the way a fashion copywriter would, not a robot.
- **Attribute tags (stretch goal):** colour, pattern, sleeve type, neckline — structured fields that power filters like "sleeveless" or "floral".

03 Model Selection

The job has two very different sub-tasks, and rather than forcing one model to do both, it helps to let each model do what it's naturally best at:

- **CLIP-style dual encoder** — trained to pull matching images and text close together in the same vector space. It doesn't "write" anything; it **recognises**, which is exactly what category placement needs: compare the photo's embedding against every category label and pick the nearest one, even for categories it has barely seen before (zero-shot).

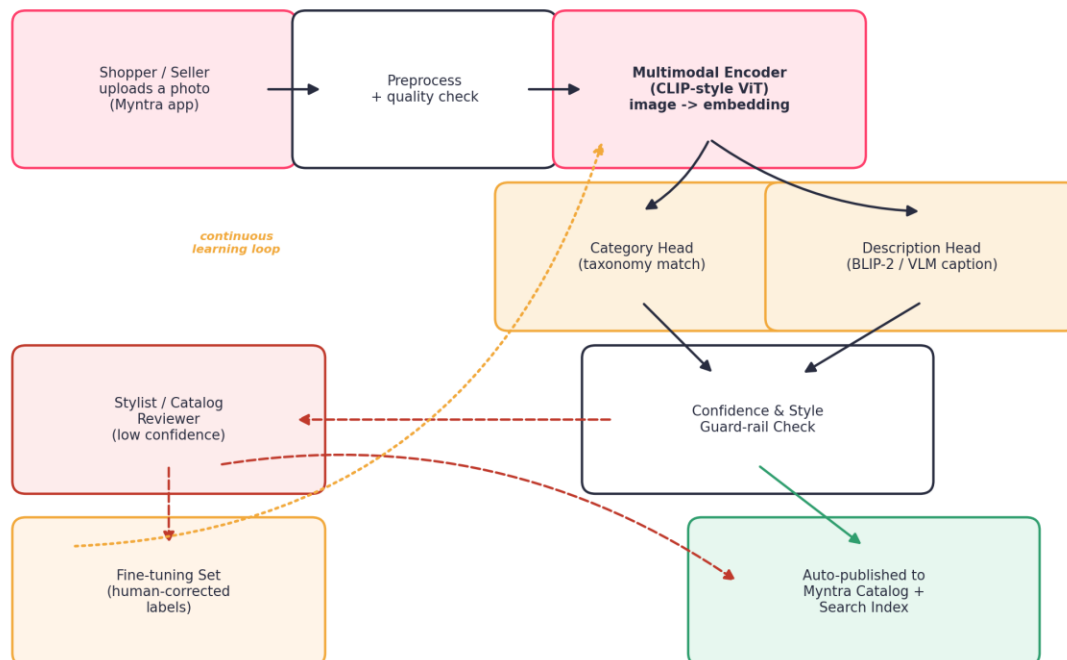
- **BLIP-2 / a vision-language model (LLaVA, GPT-4V-class)** — trained to look at an image and generate fluent, free-form sentences about it. This is the right tool for the **description**, because writing prose is a generation problem, not a matching problem.

Why not just one giant model for everything? A large VLM can technically do both, but a lightweight CLIP-style classifier is far cheaper and faster to run on every single one of the thousands of images uploaded daily, leaving the heavier generative model to focus its compute only on writing the description.

04 System Design — How It Actually Flows

1. Photo lands in the system the moment a seller (or AI stylist tool) uploads it through the Myntra app or seller portal.
2. A quick preprocessing and quality gate resizes the image and rejects blurry, watermarked, or duplicate shots before they ever reach the model.
3. A shared multimodal encoder turns the image into a single embedding — the numerical "fingerprint" both downstream heads will read.
4. That embedding fans out to two specialist heads at once: the category head places it on the taxonomy, and the description head drafts the copy.
5. A confidence and guard-rail check screens both outputs — flagging unusual categories, banned words, or text that doesn't actually match what's visible in the photo.
6. Anything that fails the check goes to a human catalog reviewer or stylist; everything that passes is auto-published straight to the live catalog and search index.
7. Crucially, every human correction is logged into a fine-tuning set, which periodically retrains the encoder and heads — a continuous learning loop, not a one-time model.

Pipeline (with the feedback loop that makes it self-improving):



05 Evaluation

"Good" here isn't one number — it's five different questions, each answered by a different kind of evidence:

Layer	What we check	Signal we trust
Model	Does the predicted category match a stylist-verified test set across categories like ethnic wear, footwear, and accessories?	Precision / Recall / F1, confusion matrix
Language quality	Is the generated description fluent, on-brand, and actually grounded in the photo rather than a generic template?	CLIPScore (image-text match), human 1-5 rating
Trust & safety	Does the description avoid hallucinated claims ("100% cotton" on a synthetic blend) or inappropriate content?	% flagged in manual audit, complaint rate
Business	Do AI-tagged listings actually get discovered and bought, compared with manually-tagged ones?	Search CTR, conversion rate, return rate
Operations	Can the system keep up with thousands of new SKUs uploaded every day across categories?	Latency per image, cost per 1,000 SKUs

A model can score well on category accuracy and still be a bad product if its descriptions sound robotic or invent details — which is why language quality and trust & safety are tracked as separate, non-negotiable layers, not folded into one "accuracy" metric.

06 Business Value

- **Minutes to seconds:** a manual SKU listing that takes a catalog associate several minutes can be auto-drafted instantly, which matters enormously at Myntra's scale of new arrivals every single day.
- **One voice across thousands of sellers:** instead of wildly inconsistent seller-written titles and descriptions, every listing reads with the same quality bar and tone.
- **Search that actually understands fashion:** correct categories and rich, attribute-rich text feed directly into better search filters and recommendation quality.
- **Fewer returns:** accurate, non-exaggerated descriptions (right fabric, right fit) reduce the "didn't look like the photo" return rate — a real cost centre in fashion e-commerce.

07 Challenges & Limitations

- **Visual ambiguity:** a co-ord set photographed alone can look like a single dress; fine-grained fashion categories often need more than just pixels to disambiguate.
- **Representation bias:** if training data skews toward certain skin tones, body types, or regional clothing styles, the model's descriptions and category accuracy will quietly be worse for everything else — this needs deliberate auditing, not just overall accuracy numbers.
- **Fast-moving trends:** fashion vocabulary changes constantly (a silhouette becomes "cottagecore" one season); the taxonomy and the model both need regular refreshes to stay current.
- **Multilingual catalogs:** an India-wide platform needs descriptions that work in English and Hindi (and beyond) — not just translated, but written naturally in each language.

08 Future Scope

- **Short video understanding:** extend the same encoder to 360-degree or try-on video clips, since fabric drape and movement are hard to judge from a single still photo.
- **Voice + image search:** let a shopper speak a follow-up ("same, but in mustard") on top of a photo search, combining Myntra's existing AI stylist assistant with this image pipeline.
- **Personalized description tone:** adapt the generated copy's style to the shopper segment reading it — Gen Z slang for one cohort, more classic copy for another.

Bonus Real-World Example — Myntra

Myntra (part of the Walmart-owned Flipkart Group) is a genuinely strong real-world example, because it has shipped multimodal AI in more than one form:

- **Visual search:** a shopper can photograph or screenshot an outfit they like, and Myntra's image-search engine — built with vision partners like ViSenze/Rezolve AI — finds visually and stylistically similar catalog items, with reported image-search traffic growth of roughly 35% year-over-year.
- **"Shop the Look" and "View Similar":** the same visual-matching technology automatically surfaces every individual item worn in a model photo, and complementary pieces, turning one photo into a whole shoppable outfit.
- **AI Fashion Assistant:** built on Azure OpenAI Service, this lets shoppers type natural-language requests like "something for a beach wedding" and get a complete, multi-category outfit back — the text-generation half of the same multimodal idea explored in this assignment.

09 A Personal Note

What stands out to me while designing this is that the hard part isn't getting a model to recognise "this is a kurta" — that's almost solved. The hard part is everything around it: catching the 2% of cases where the model is confidently wrong, keeping the system fair across India's huge diversity of clothing styles, and building the feedback loop so today's mistakes become tomorrow's training data. That's the piece I found most interesting to design.

— Akash C Shetty